	$\omega = 20.73 = 21 \text{ rad s}^{-1}$	A1
3ai	Temperature at which all substances have a <u>minimum</u> (internal/kinetic) energy.	B1
3aii	<u>Readings</u> from <u>constant-volume</u> gas thermometer (accept constant-pressure) are plotted on a pressure-temperature graph. By <u>extrapolating the graph</u>, the temperature at which pressure is zero is -273.15 °C. This temperature is the same (when experiment is repeated) for different gases OR this temperature does not depend on the property of any particular substance.	B1 B1 B1
3b	Before the tap is opened, for cylinder B, pV = nRT $(1.2 \times 10^5)(2.0 \times 10^{-2}) = n_B (8.31)(273.15 + 37)$ $n_B = 0.931$ mol (Cylinders are insulated, there is no heat exchange with their surroundings, Q = 0. Since total volume is fixed, W = 0. For a system consisting of the two cylinders, using 1st Law of Thermodynamics, this means $\Delta U = 0$, thus for ideal gas, this means $\Delta T = 0$) Final temperature is 37°C After tap is opened, using pV = nRT on for both cylinders $p(V_A + V_B) = (n_A + n_B)RT$ $p(2 \times 2.0 \times 10^{-2}) = (1.2 + 0.931)(8.31)(273.15 + 37)$ $p = 1.4 \times 10^5$ Pa	 C1 C1 C1 A1
4ai	Frequency of the system when it is oscillation freely	B1